

Report

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HW 3, Programming part

Output from NaiveBayes.py:

```
● (base) anshpachauri@Anshs-MacBook-Pro NaiveBayes % python3 NaiveBayes.py
Total Sentences correctly: 775
Predicted correctly: 580
Accuracy: 74.83871%
```

Output from NaiveBayes_improved.py

```
● (base) anshpachauri@Anshs-MacBook-Pro NaiveBayes % python3 NaiveBayes_improved.py
Total Sentences correctly: 775
Predicted correctly: 687
Accuracy: 88.64516%
```

In the improved version of the Naïve Bayes classifier, I addressed the issue of zero probabilities that led to infinite negative log values during prediction by incorporating Laplace smoothing. Specifically, while estimating the conditional probabilities $p(x[d] = 1 | Y = c)$, I added a pseudo count of 1 to the numerator and 2 to the denominator for each binary feature. This adjustment ensured that no probability was exactly 0 or 1, thereby stabilizing the log probability computations. As a result, the improved classifier achieved an accuracy of approximately 88.65%, a substantial improvement over the baseline accuracy of around 74.84%.